ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Scalable and Reliable Services

SRS

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The course

- **Classroom lessons**: 64 hours - **Language**: English/Italian

- **Main objective**

  Modern digital services must satisfy various requirements in terms of **performance**, **scalability**, **reliability** and security. We analyze the whole lifecycle of high-quality services, from design to deployment, testing and operation through genAI and agentic AI. In each phase, we aim to provide participants with the ability to gather and analyze data, to identify sources of outages and critical dependencies, and to avoid bottlenecks and single points of failure. Finally, we expect participants to be prepared to evaluate and discuss alternative solutions and justify right investments

- **Prerequisites**
  - Fundamental courses characterizing any Bachelor in Computer Science or Computer Engineering: systems, networking, software
  - Some knowledge about distributed systems, cloud, software engineering is preferred
  - Some knowledge about **genAI** is preferred as well

Calendar

- 5 hours per week, every Monday and Tuesday, but:
  - March 23 and 24: Graduation days
  - April 6: Easter holidays

- Monday, 9:15-11:45, Room 2.7A
- Tuesday, **11:30**-13:30, Room 5.5

If everything goes well, teaching period ends by May 26

Exact schedule is subject to variations independently of my will.
*Please, check notes and advices on **virtuale.unibo.it***

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Exams (*formal dates*)

- **June 15** – Room 5.2, h. 9
- **July 6** – Room 5.2, h. 9
- **July 23** – Room 5.2, h. 9

From September 2026, on demand

Past agenda (1/2)

SRS

Introduction

- Cloud services and cloud provider architectures
- High-quality services: performance, scalability, reliability (and security)
- Lifecycle of high-quality services

Service deployment

- Secure and robust software production
- Secure and robust DevOps
- Labs

Architectures for scalable and reliable services

- Sources of outages and critical dependencies
- Bottlenecks and single points of failure
- Design
- Maturity assessment
- Architectural principles for reliability and scalability
- Migration to cloud
- Critical data management
- Strategy, Governance, Management

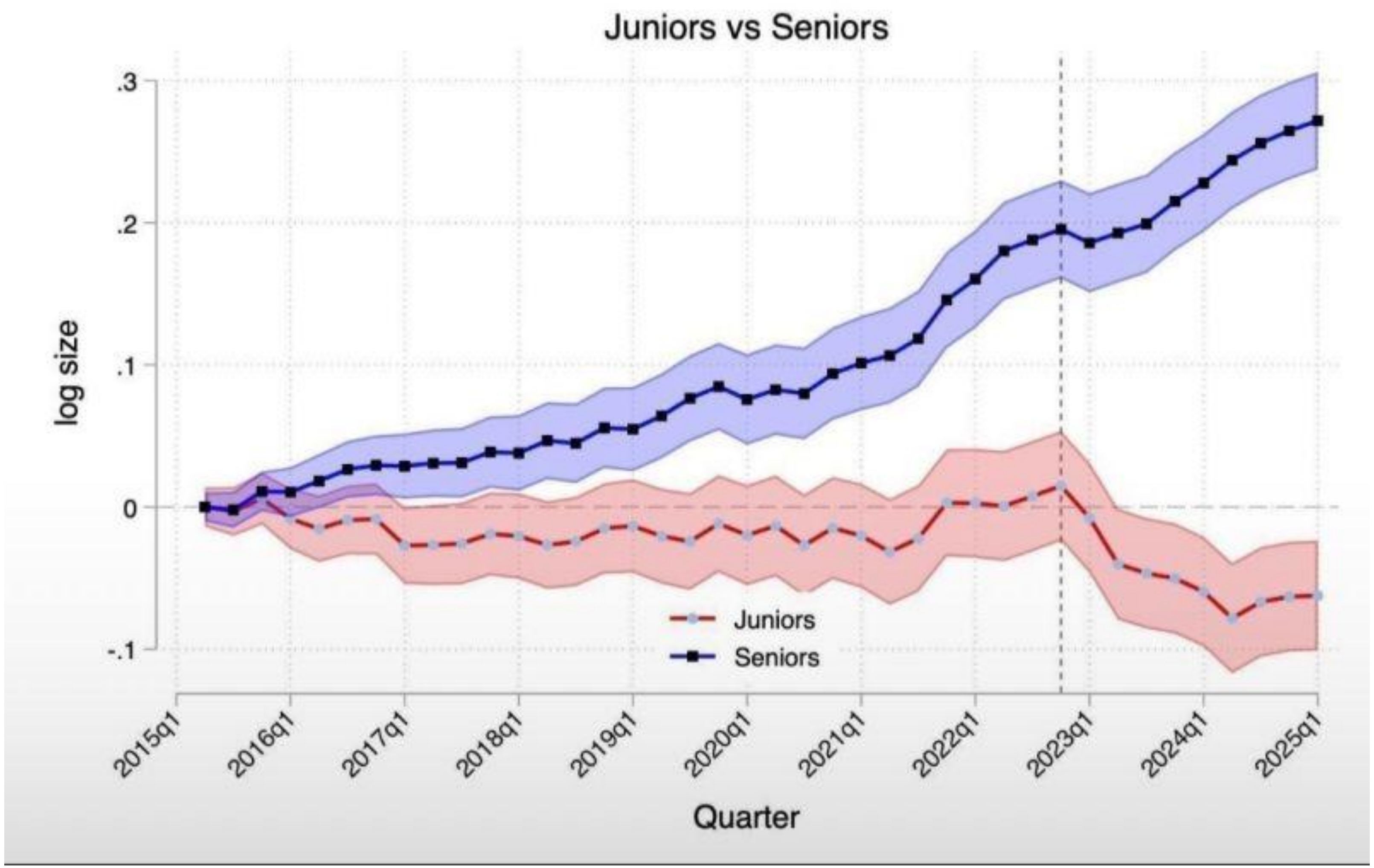
Past agenda (2/2)

Tests
- Benchmarking
- Performance testing
- Stress testing

Scalable and reliable organizations
- Roles and responsibilities
- Leaders and managers
- Putting all together
- Capacity planning
- Understanding and managing complexity
- Managing changes

Issues, incident and crises
- Performance monitoring
- Warning signs in operation
- Incidents management
- Crisis management

Effects on hiring in one year



Harvard monitored 285,000 companies and confirmed the *AI hiring apocalypse* we suspected:

- **Junior** roles have decreased by 23%
- **Senior** roles have increased by 14%.Harvard
- Before AI: 1 senior + 3 juniors = 4-person team
- After AI: 1 senior + Claude/Gemini = same productivity

We are creating a generation of experts without apprentices

Purposes *vs* Task [Jen-Hsun Huang / Jensen Huang]

https://www.youtube.com/shorts/MscLPdZVMA0

Task of a software engineer:

- "Coding"

Purposes of a software engineer:

- "To solve known problems"
- "To find new problems to solve"

Agenda

SRS2026

Introduction

- Cloud services and cloud provider architectures
- High-quality services: performance, scalability, reliability (and security)
- Lifecycle of high-quality services

Service deployment through genAI and agentic AI

- Secure and robust software production, Labs
- Secure and robust DevOps, Labs

Architectures for scalable and reliable services through genAI and agentic AI

- Architectural principles for reliability and scalability
- Sources of outages and critical dependencies
- Bottlenecks and single points of failure, Labs
- Critical data management, Labs
- Strategy, Governance and Management for scalable and reliable services, Labs

Tests

- Benchmarking
- Performance testing
- Stress testing

Incident and crisis management (TBD)

- Performance monitoring
- Warning signs in operation

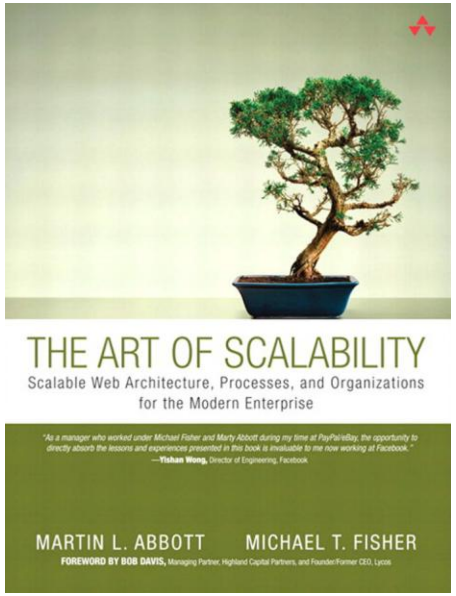
Types of exam 2026

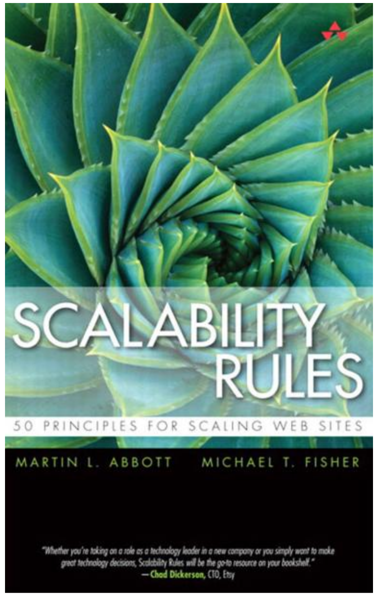
Students that are oriented to take this course as a priority and to pass the exam **formally on June 15, 2026 (actually, May)** have one option:

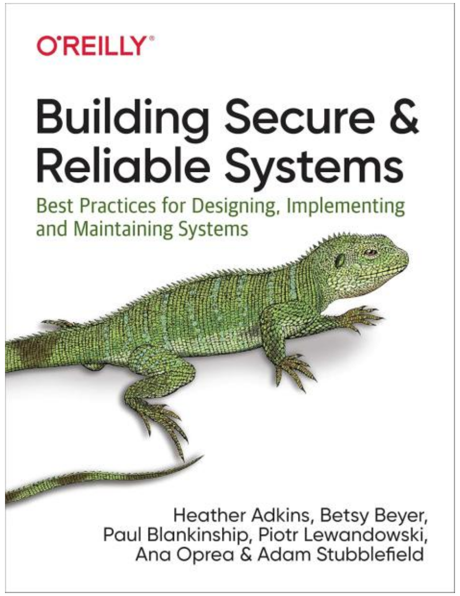
A. **Participation to our experimentation 2026 during the labs** (*typically on Tuesday*)

From **July 2026 onwards**, there is only one possibility:

B. **Traditional oral exam based on books**







Project Type A

- Google GCP and Gemini platform
- Other AI agentic tools
- Novel 2026 requirements:
  - Service requirements
  - Budgeting
  - ROI
  - Reliability test
  - Use of Agentic AI as far as you can (with guardrails)
  - Data management (and data cleaning)

Plan 2026

- 16, 17, 23, 24 February: SRS, Data centers, Cloud, genAI and agenticAI

- 2 March: Tools and frameworks for genAI and agenticAI
- **3 March**: SRS and project assignment
- 9, 16, (no 23), 30 March: SRS lessons
- 10, 17, 24, 31 March: exercises (30m theory + 90m exercises)

Easter period: *First check*

- **April**: 6 lessons and exercises (mix TBD)

May 1st period: *Second check*

- **May**: 8 lessons and exercises (mix TBD)

May 26th: end of the course with first possible delivery of projects

June 15th: second delivery of projects

Possible final objective: *training attitude when there is a serious problem*

Incident and crisis management (TBD)

- Performance monitoring
- Warning signs in operation
- Act and solve

No discussions. It only matters:

1) Who knows how to get there and limit the problem?
2) Who knows how to fix the problem?
3) Who takes the responsibility for fixing?

True seniority is not having all the answers. It is feeling unsure when you do not have "trained" people that can answer.

Hire problem solvers! A resume does not fix bugs. **Attitude does, training to exercise attitude does even better**.

Outcome is the only metrics that matters

Slides and other material

https://virtuale.unibo.it/

Integration with other activities

- The SRS activities may be preparatory for the Project Work (3-4 CFUs) that is contemplated in some *Master Laurea* courses
- They are presented and evaluated at different times and with specific assessments
- The SRS activities + project work may be preparatory for thesis


Questions